Poster No.P07

Integrated Virtual Screening of Natural Product-Like A_{2A} Receptor Antagonist Candidates for Parkinson's Disease

Adisorn Sirichotejirakul[1,2], Pornchai Rojsitthisak[2,3], and Worathat Thitikornpong[2,3*]

[1]Pharmaceutical Sciences and Technology Program, Faculty of Pharmaceutical Sciences, Chulalongkorn University, Bangkok 10330, Thailand
[2]Center of Excellence in Natural Products for Ageing and Chronic Diseases, Chulalongkorn University, Bangkok 10330, Thailand
[3]Faculty of Pharmaceutical Sciences, Chulalongkorn University, Bangkok 10330, Thailand
* Corresponding author: worathat.t@pharm.chula.ac.th

1. INTRODUCTION

Parkinson's disease is characterized by progressive dopaminergic neurodegeneration leading to motor dysfunction. Current first-line therapies such as levodopa are limited by long-term motor complications including dyskinesia, highlighting the need for complementary non-dopaminergic strategies. The adenosine A_{2A} receptor is a promising alternative target, as its pharmacological blockade modulates striatal indirect pathway signaling and supports dopamine-related motor function.[1] This study developed an AI-integrated consensus virtual screening workflow to prioritize natural product-like A_{2A} receptor antagonist candidates by integrating machine learning (ML)-based activity prediction, GNINA deep-learning docking, and drug-likeness assessment.[2,3]

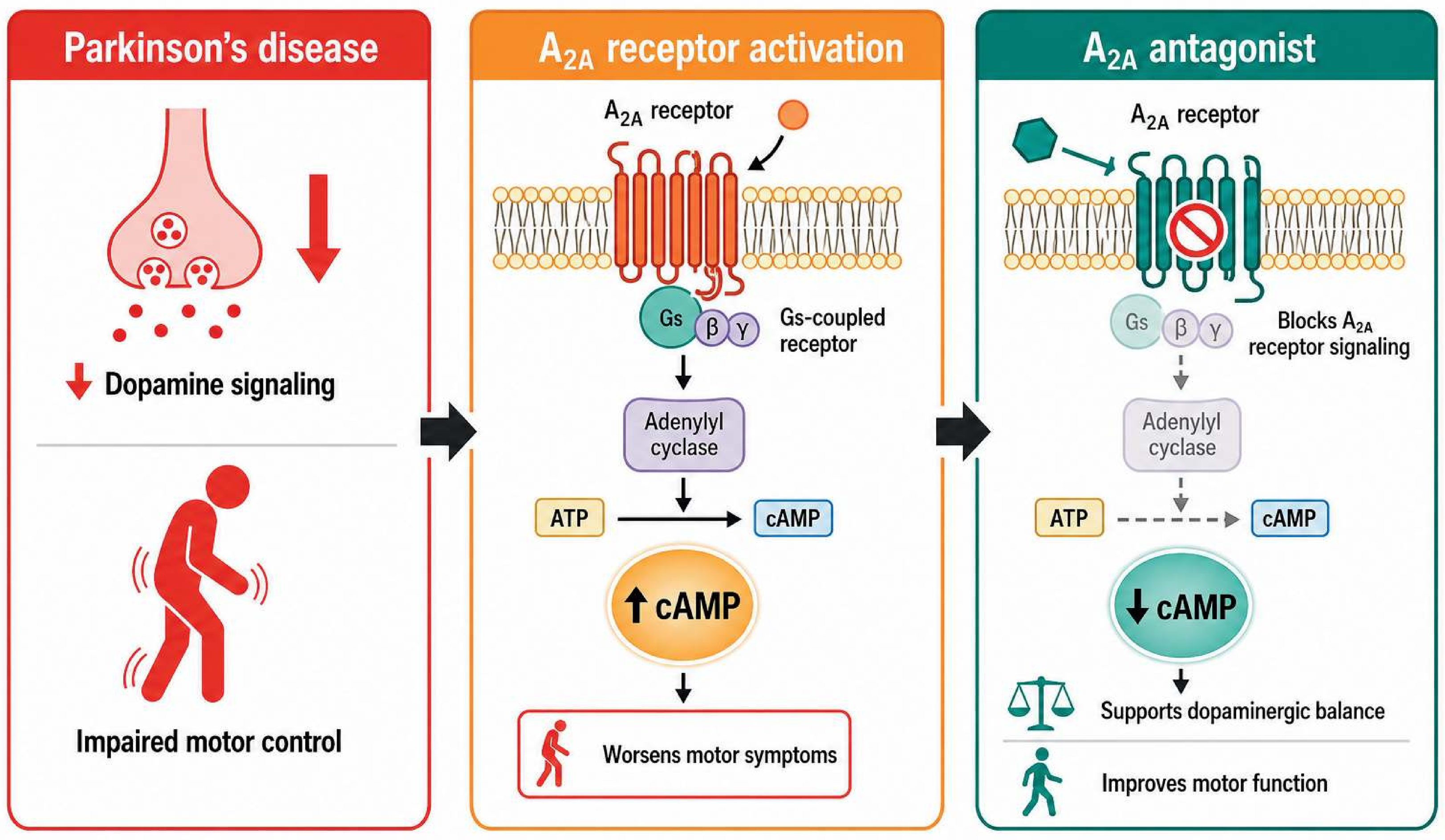


Figure 1. Rationale for targeting the A_{2A} receptor in Parkinson's disease. A_{2A} receptor blockade reduces cAMP-mediated signaling in striatal neurons, supporting dopamine-related motor function.[1]

2. OBJECTIVES

- To develop an AI-integrated consensus virtual screening workflow for identifying natural product-like A_{2A} receptor antagonist candidates.
- To evaluate candidate compounds using three evidence streams: machine learning-based activity prediction, GNINA deep-learning docking, and drug-likeness assessment.
- To prioritize Tier 1 candidates for future A_{2A} receptor binding and functional validation.

3. METHODOLOGY

An AI-integrated workflow combined ML activity prediction, GNINA docking, drug-likeness profiling, and consensus filtering to prioritize high-confidence A_{2A} receptor antagonist candidates.

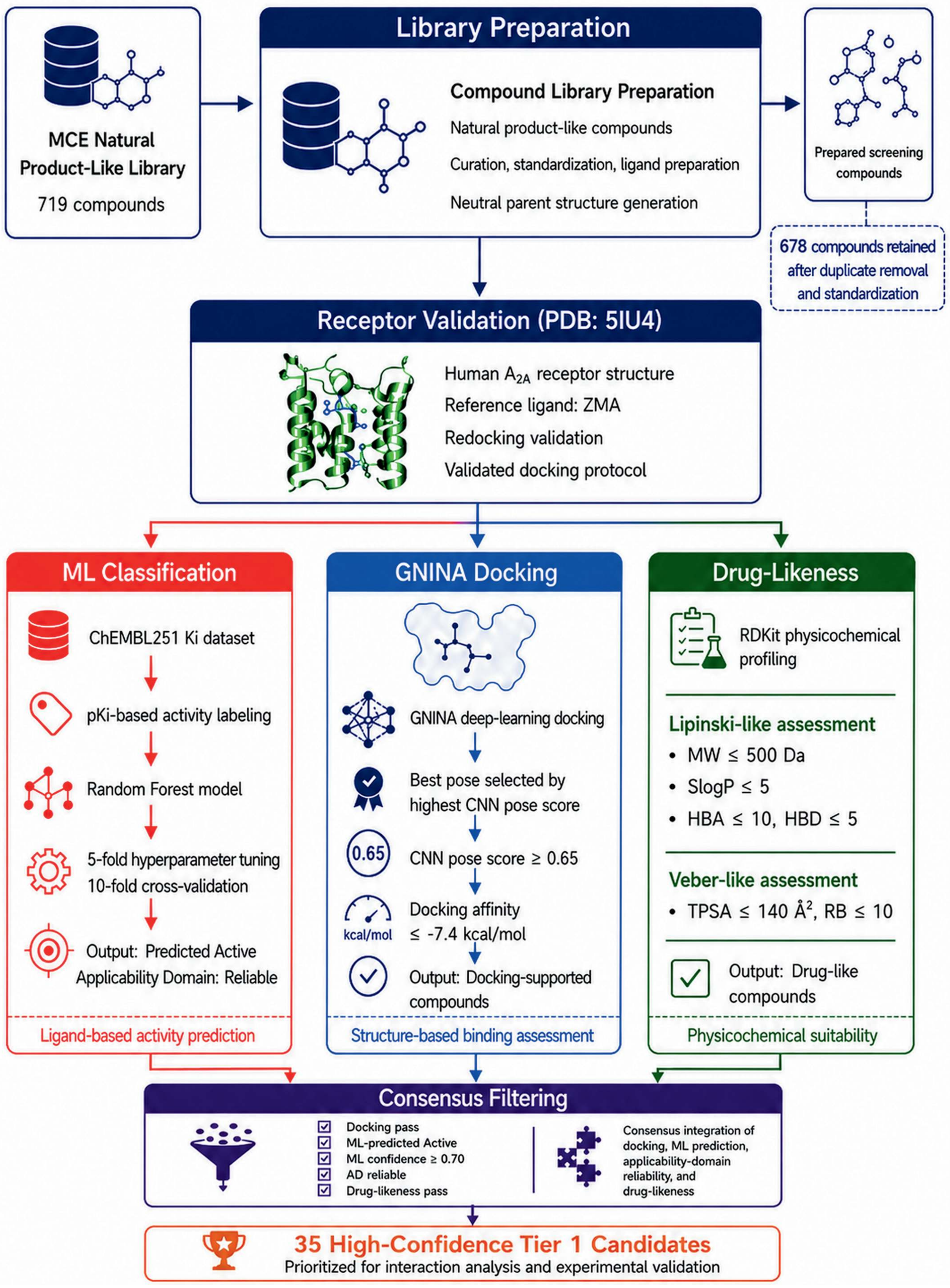
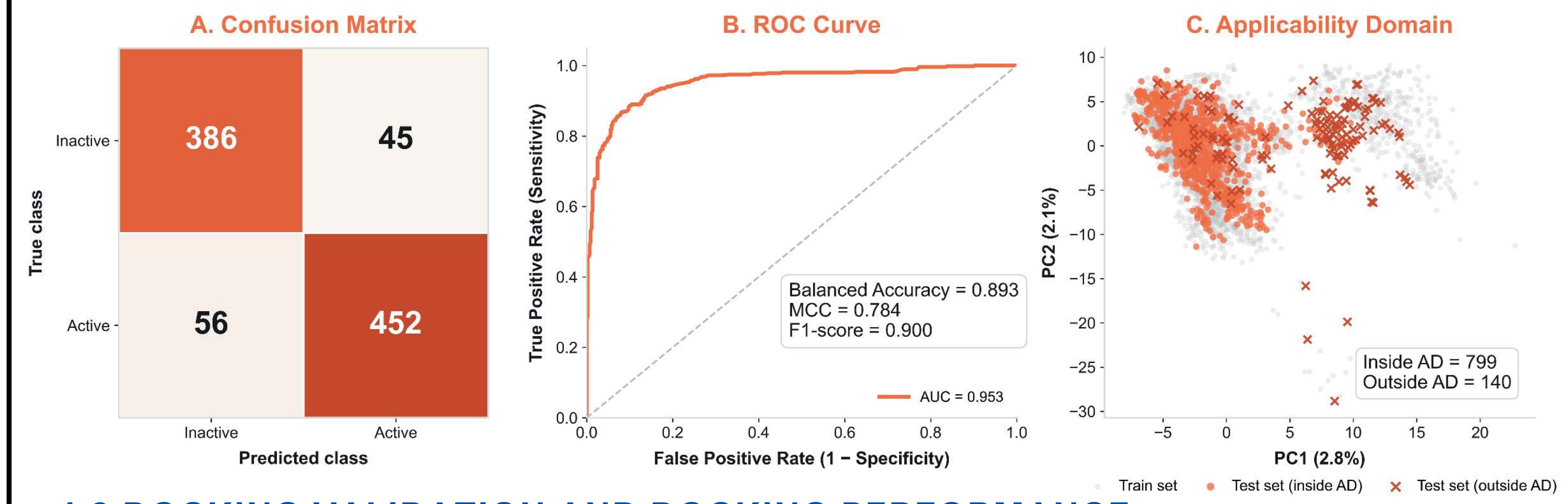


Figure 2. AI-integrated consensus virtual screening workflow. The MCE Natural Product-Like Library (719 compounds) was curated and standardized, yielding 678 un-ionized parent structures for downstream screening. Compounds were evaluated using ML activity prediction, GNINA docking, drug-likeness profiling, and consensus filtering.[2-4]

4. RESULTS AND DISCUSSION

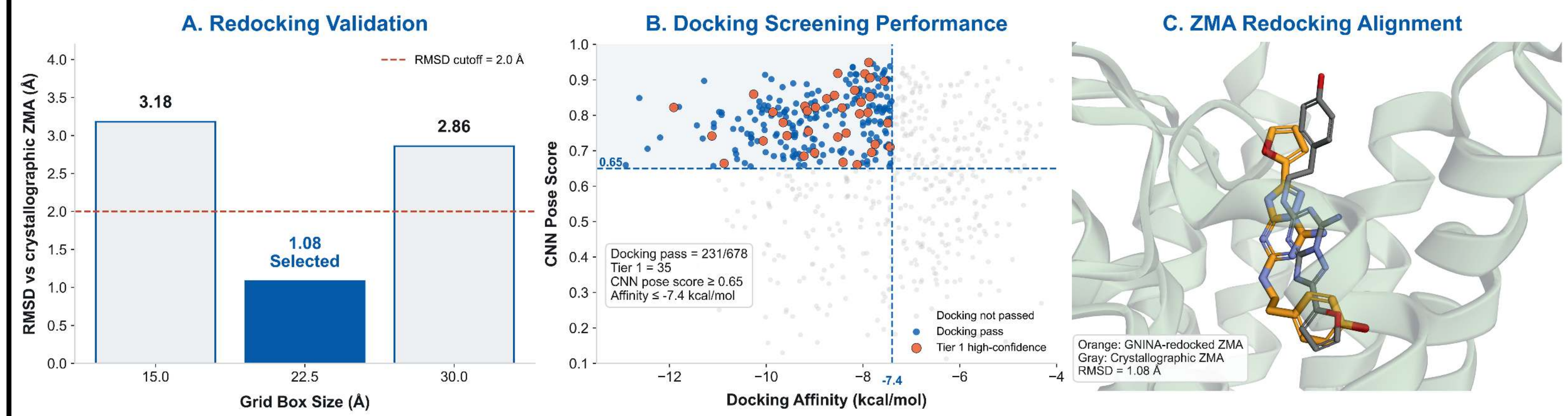
4.1 MACHINE LEARNING MODEL VALIDATION

The final Random Forest model demonstrated strong discriminative performance (balanced accuracy = 0.893, MCC = 0.784, F1-score = 0.900, ROC-AUC = 0.953) and was applied within the defined applicability domain.



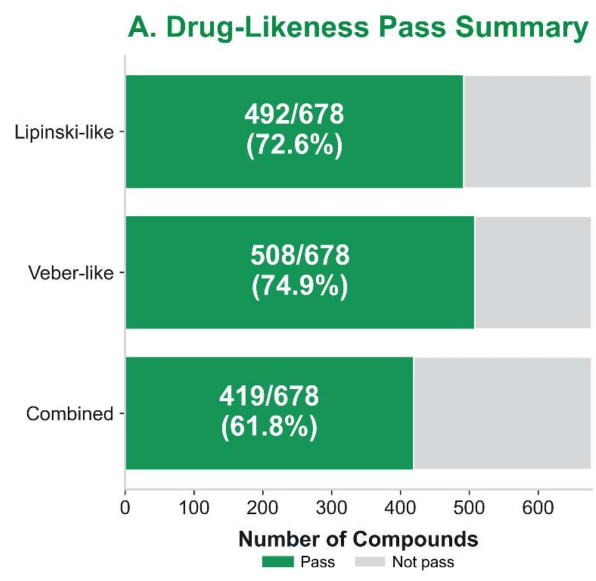
4.2 DOCKING VALIDATION AND DOCKING PERFORMANCE

Redocking of the co-crystallized ligand ZMA at the 22.5 Å × 22.5 Å × 22.5 Å grid box yielded RMSD = 1.08 Å (threshold < 2.0 Å), confirming protocol validity.[3,4] GNINA screening of 678 compounds applied CNN pose score ⩾ 0.65 (pose-quality threshold) and docking affinity ⩽ −7.4 kcal/mol (approximately the median library affinity), yielding 231 docking-supported compounds.



4.3 DRUG-LIKENESS PROFILING

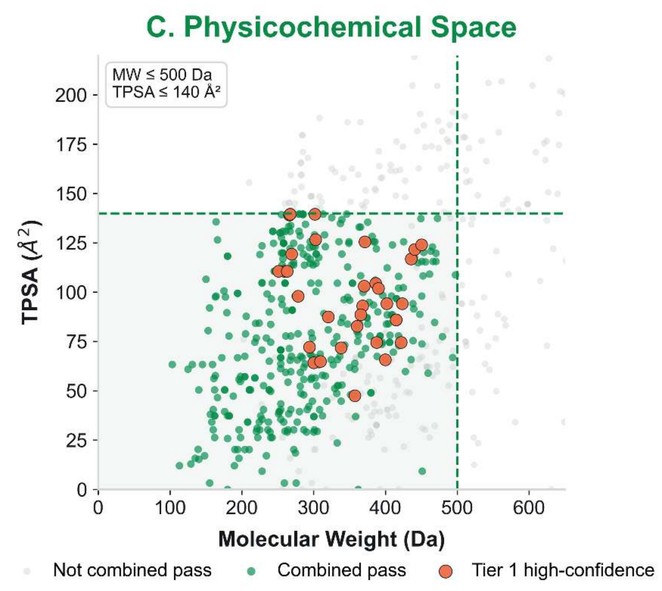
Drug-likeness filtering retained compounds satisfying Lipinski-like (MW ⩽ 500 Da, SlogP ⩽ 5, HBA ⩽ 10, HBD ⩽ 5) and Veber-like (TPSA ⩽ 140 Å², RB ⩽ 10) physicochemical criteria, yielding 419 of 678 screened library compounds (61.8%) for consensus integration.[5,6]



B. Tier 1 Property Profile

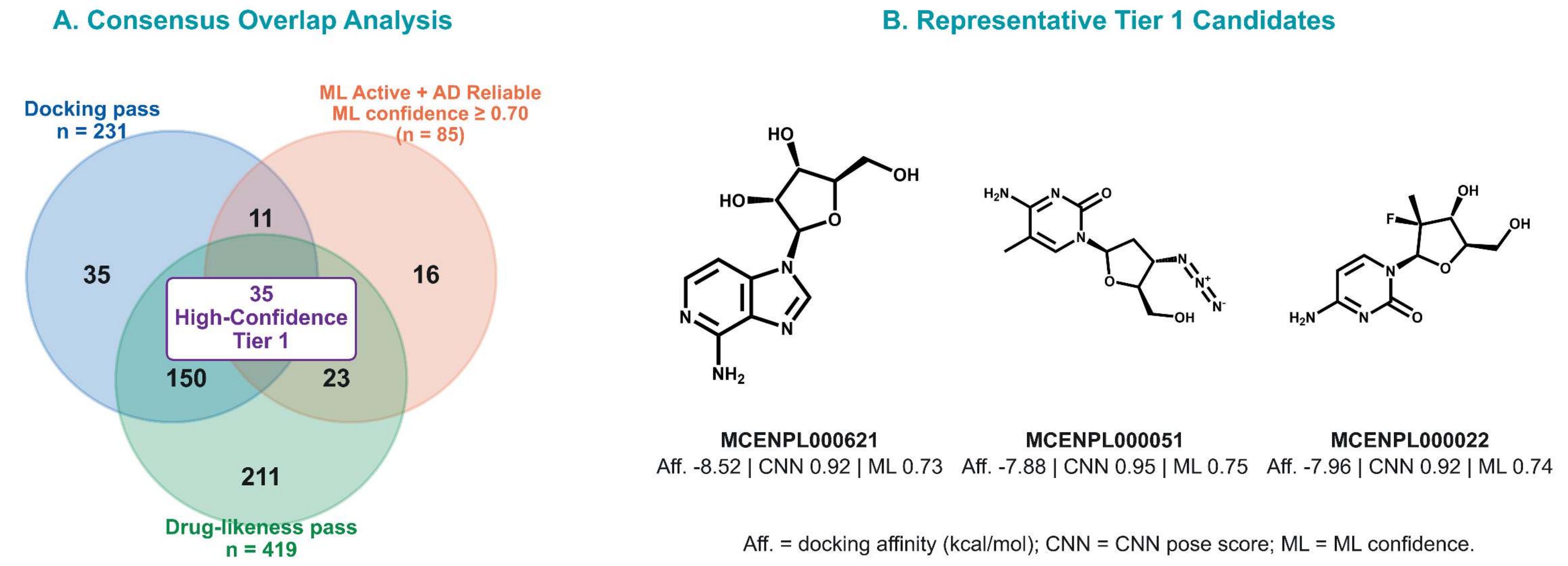
Property	Median	Range	Criterion
MW	360.4	251.2–450.5	≤ 500 Da
SlogP	2.1	-2.0–4.6	≤ 5
TPSA	98.0	47.6–139.5	≤ 140 Å²
RB	3	2–10	≤ 10
HBD	2	1–4	≤ 5
HBA	6	3–9	≤ 10

All Tier 1 candidates passed Lipinski-like and Veber-like criteria (n = 35).



4.4 CONSENSUS PRIORITIZATION AND REPRESENTATIVE TIER 1 CANDIDATES

Consensus filtering prioritized 35 high-confidence Tier 1 candidates satisfying docking pass, ML confidence ⩾ 0.70, applicability domain (AD) reliability, and drug-likeness compliance. Three representative candidates were selected based on the highest consensus scores and structural diversity.



Aff. = docking affinity (kcal/mol); CNN = CNN pose score; ML = ML confidence.

5. CONCLUSION

- An AI-integrated consensus virtual screening workflow was developed for natural product-like A_{2A} receptor antagonist candidate prioritization.
- The workflow integrated ML-based activity prediction, GNINA deep-learning docking, applicability domain assessment, and drug-likeness profiling.
- Thirty-five high-confidence Tier 1 candidates were prioritized for further experimental validation using A_{2A} receptor binding and functional assays.

Translational Tool: A2A-iCAP Platform

The A2A-iCAP platform enables compound input, activity prediction, docking-informed interpretation, and consensus-based prioritization.

6. ACKNOWLEDGEMENTS

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REFERENCES
1.Korkutata M, Agrawal L, Lazarus M. Allosteric modulation of adenosine A2A receptors as a new therapeutic avenue. Int J Mol Sci. 2022;23(4):2101. doi:10.3390/ijms23042101.
2.Zhou G, Rusnac DV, Park H, Canzani D, Nguyen HM, Stewart L, et al. An artificial intelligence accelerated virtual screening platform for drug discovery. Nat Commun. 2024;15:7761. doi:10.1038/s41467-024-52061-7.
3.Buccheri R, Rescifina A. High-throughput, high-quality: benchmarking GNINA and AutoDock Vina for precision virtual screening workflow. Molecules. 2025;30(16):3361. doi:10.3390/molecules30163361.
4.Ivanova L, Karelson M. The impact of software used and the type of target protein on molecular docking accuracy. Molecules. 2022;27(24):9041. doi:10.3390/molecules27249041.
5.Pollastri MP. Overview on the Rule of Five. Curr Protoc Pharmacol. 2010;49:9.12.1-9.12.8. doi:10.1002/0471141755.ph0912s49.
6.Veber DF, Johnson SR, Cheng HY, Smith BR, Ward KW, Kopple KD. Molecular properties that influence the oral bioavailability of drug candidates. J Med Chem. 2002;45(12):2615-2623. doi:10.1021/jm020017n.